

TaSain Thomas

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SUMMARY

AI/ML Engineer bridging published research and production systems. Translated peer-reviewed virtual-anthropometry literature into a secure VLM evaluation pipeline, extending pose-to-ratio methodology with GPT-as-Judge reasoning and LoRA fine-tuning - achieving 89% BMI estimation accuracy at 95% training cost reduction. Industry 4.0 automation background delivering 33% throughput gains at a major automotive manufacturer. Seeking to apply ML validation, research translation, and adversarial robustness expertise to safety-critical AI systems.

EDUCATION

Wayne State University, James and Patricia Anderson College of Engineering

Bachelor of Science, Computer Science

• GPA: 3.4/4.0

May 2026

Detroit, MI

CORE COMPETENCIES

CORE COMPETENCIES: ML Validation, CI/CD, Red Teaming, Model Monitoring, LoRA/VLM Fine-Tuning, RAG--

, Simulation-Based Evaluation, Uncertainty Quantification, Kubernetes, Docker, Python, PyTorch, C++, Adversarial Robustness

TECHNICAL SKILLS: Python, C++, JavaScript, SQL, Java, Dart, PyTorch, TensorFlow, Hugging Face, LangChain-, LoRA/QLoRA, RAG, Vector Databases, Transformer Architectures, Kubernetes, Docker, CI/CD, Model Serving & Monitoring, Helm, Terraform, GPU Passthrough, AWS/Azure, DigitalOcean, Nginx, Cloudflare, Proxmox, systemd, Red Teaming, Prompt Injection Defense, Zero Trust, Adversarial Robustness, Kali Linux

PROFESSIONAL EXPERIENCE

Magna EV Structures

Industry 4.0 / Automation Intern (Rotational)

Sep 2023 - Sep 2024

St. Clair, MI

- Delivered four automation projects on schedule by integrating real-time production data pipelines via custom API development.
- Improved throughput 33% by implementing Vuforia Model Targets for vision-guided assembly verification.
- Built C++/SQLite3 worker-to-station matching engine with evaluateFit() scoring algorithm, reducing assignment time by 40%.
- Reduced debug cycle time 25% by developing custom C# validation logic that automated error handling and root-cause identification.

ENGINEERING PROJECTS

ConformalBMI: Uncertainty-Aware Body Mass Estimation from Images [GitHub](#)

May 2026 - Aug 2026

- Translated published BMI-estimation research (Digital Scale; Sui et al.; Manichand et al.; Jin et al.) into a working SE-DenseNet121 (PyTorch) estimator on Celeb-FBI (7,208 images), achieving 2.81 BMI-point MAE (11.56% MAPE)
- Trained on Celeb-FBI (7,208 real images, 6,174 labeled); final performance of 2.81 BMI-point MAE and 11.56% MAPE on held-out test data.
- Layered an LLM-as-judge reasoning module (GPT-4.1-Nano) to flag high-uncertainty and extreme-BMI tail cases, converting a point estimate into a calibrated, explainable prediction.
- Extended the model with split conformal prediction for distribution-free uncertainty -- 90% prediction intervals ($q = 6.44$) with empirically verified 90.4 coverage on unseen samples.

MLOps Infrastructure & AI Agent Engineering

Mar 2026 - Present

- Architected auto-scaling Kubernetes inference cluster with GPU passthrough serving LLMs at 99.9% uptime, with model monitoring and real-time incident response.
- Embedded red-team security testing (Kali Linux) into CI/CD, patching prompt injection vulnerabilities pre-production.

SaaS Booking & Payment Platform [GitHub](#)

Mar 2026 - Present

- Engineered full-stack CRM (Python/JavaScript) with Stripe payments and Telegram Bot API; deployed on DigitalOcean, Nginx, systemd, Cloudflare CI/CD at 24/7 uptime.

Real-Time Mobile Collaboration Platform

Jan 2026 - May 2026

- Developed cross-platform (Flutter/Dart) collaborative workspace integrating ZegoCloud real-time video and RevenueCat subscription monetization; shipped to production for client deployment.